

Session Guide

One page per week. Each page carries everything that session needs — which pages the students are on, what to prepare, the clock, what counts as done, and the teaching notes — so a session can be led without reading the whole companion first.

WK	DATE	TITLE	NOTEBOOK	GUIDE	COMPANION
1	Sep 16	Build and drive	pp. 6–7	pp. 1–2	pp. 2, 6–7
2	Sep 23	Find your grip	pp. 8–9	— no new code	p. 4
3	Sep 30	Speed or shove?	pp. 10–11	— no new code	pp. 5, 8
4	Oct 7	Choose your fighter	pp. 12–13	p. 2	pp. 2, 7
5	Oct 14	Make it move	pp. 14–15	p. 2	p. 7
6	Oct 21	Repeat and turn	pp. 16–17	p. 3	p. 8
7	Oct 28	Feel a hit	pp. 18–19	pp. 4, 7	p. 9
8	Nov 4	Can it stay in?	pp. 20–21	p. 5	p. 10
9	Nov 11	Find the opponent	pp. 22–23	pp. 5, 8	p. 11
10	Nov 18	Choose what happens	pp. 24–25	pp. 6, 8, 9, 10	pp. 12–14
11	Dec 2	Make one improvement	pp. 26–27	— no new code	p. 21 (concept notes) · p. 14
12	Dec 9	Tournament day	pp. 28–29	—	p. 15

Three things this guide does not invent. Where a source document has no answer, the page shows a blank rather than a guess. **Watch for** is filled in only on the weeks the companion actually documents a failure mode — the rest are ruled lines for the lead mentor to complete after the pilot. **Week 11** has no dedicated mentor page and conflicts with the public parent syllabus. **Week 12** has no clock in the companion; the one shown is a proposal. All three are marked on their pages.

Build and drive

Mission: Build Bevel Bot. Make it drive and stop.

STUDENT NOTEBOOK

pp. 6–7 • first program p. 5

PROGRAMMING GUIDE

pp. 1–2

MENTOR COMPANION

pp. 2, 6–7

BEFORE STUDENTS ARRIVE

Bevel Bot pictures ready; sort parts into build sections; test a single motor on A.

PORTS TONIGHT

A = Bevel Bot motor (weeks 1–2)

THE HOUR — 50 PLANNED, 10 HELD FOR RECOVERY

5 welcome • 25 build • 12 program & test • 3 evidence • 5 clean up

Teaching notes

- Students build the program themselves — no prewritten code is supplied (M p. 1).
- Measure travel from the same front edge every time; they mark that edge on the sketch (N p. 7).
- Guide p. 1 is connect-and-create; Guide p. 2 is the sequence stack.

DONE TONIGHT

Build Bevel Bot; students create the A-motor program; one measured drive/stop.

IF TIME REMAINS

Three 100 cm trials.

WATCH FOR

Not documented in the companion for this week — add what you see in the room.

OPEN ITEM

The three-trial error table on N p. 6 is marked *Optional* and repeats Week 5's core exercise. Decision pending — see the note on the Week 5 page.

Find your grip

Mission: Find the setup that slides at the steepest angle.

STUDENT NOTEBOOK

pp. 8–9

PROGRAMMING GUIDE

— no new code

MENTOR COMPANION

p. 4

BEFORE STUDENTS ARRIVE

Checked ramp and restraint, angle gauge, catch area. Keep team Bevel Bots intact.

PORTS TONIGHT

A = Bevel Bot motor

THE HOUR — 50 PLANNED, 10 HELD FOR RECOVERY

5 arrive • 7 predict • 25 build & test • 8 evidence • 5 pack up

Teaching notes

- ▶ Restrain every driven axle with a pinned beam against an exposed axle gear; check by hand with power off (M p. 4).
- ▶ Keep mass, weight placement, orientation, board surface and how the angle is read all fixed. Change one thing.
- ▶ Run 3 trials per setup. A roll or a tip is **invalid** for slide-angle analysis — record it, fix the setup, repeat.

DONE TONIGHT

Two grip setups; three valid slides each; one brace comparison.

IF TIME REMAINS

Powered pushing demonstration.

WATCH FOR

Pilot acceptance (M p. 4): both wheels stay locked; the restraint never touches the board; the robot slides without falling; the angle reads consistently. No powered stall used as a brake.

Speed or shove?

Mission: Compare three gear setups. Pick your trade-off.

STUDENT NOTEBOOK
pp. 10–11

PROGRAMMING GUIDE
— no new code

MENTOR COMPANION
pp. 5, 8

BEFORE STUDENTS ARRIVE

Three separate labelled station models built before class — A 24→24 (1:1), B 12→36 (3:1), third per M p. 5. Safe 2 m lanes. Keep Bevel Bots intact.

PORTS TONIGHT

Label the actual station ports this week (M p. 2)

THE HOUR — 50 PLANNED, 10 HELD FOR RECOVERY

5 arrive • 7 predict • 25 build & test • 8 evidence • 5 pack up

Teaching notes

- Each team runs **one** assigned station, then copies the other two stations' results and names the source.
- Tug test is mentor-led: drive 0.5 s, then stop. Swap sides and repeat (M p. 8).
- Robots can differ in more than gearing — have teams note those differences before comparing.

DONE TONIGHT

Assigned station: three timed runs; share data; observe the A/B tug.

IF TIME REMAINS

Speed calculation.

WATCH FOR

Not documented in the companion for this week — add what you see in the room.

Choose your fighter

Mission: Build a two-motor base. Choose one useful improvement.

STUDENT NOTEBOOK
pp. 12–13

PROGRAMMING GUIDE
p. 2

MENTOR COMPANION
pp. 2, 7

BEFORE STUDENTS ARRIVE

Two-motor Driving Base pictures; sort parts and stage subassemblies for teams needing help. Reuse each team's kit.

PORTS TONIGHT

B/C = left/right drives from tonight — not A

THE HOUR — 50 PLANNED, 10 HELD FOR RECOVERY

5 compare ideas • 30 build • 10 program & check • 5 clean up

Teaching notes

- Save a Bevel Bot photo and program before dismantling anything.
- Keep the standard drivetrain tonight. Week 4 uses N = 1 for a short direction/stop check only (M p. 7).
- Start inside 30 × 30 cm and leave room for the sensors that arrive in Week 7.

DONE TONIGHT

Build the standard B/C base; forward/stop; compare two small ideas.

IF TIME REMAINS

One improvement and a size check.

WATCH FOR

The drive program changes with the ports. A base built tonight will not run the Week 1 A-motor program — rebuild the stack from Guide p. 2.

Make it move

Mission: Turn a motor command into a measured journey.

STUDENT NOTEBOOK pp. 14–15 • motion card p. 32	PROGRAMMING GUIDE p. 2	MENTOR COMPANION p. 7
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BEFORE STUDENTS ARRIVE Nothing new — B/C bases from Week 4.	PORTS TONIGHT B/C drives
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THE HOUR — 50 PLANNED, 10 HELD FOR RECOVERY
5 arrive • 7 predict • 25 build & test • 8 evidence • 5 pack up

Teaching notes

- ▶ Week 5 calculates N from measured wheel size and the actual gear ratio (M p. 7) — this is the calculation week, not a tuning week.
- ▶ Students use the motion card on N p. 32, then the B/C Movement stack in Guide p. 2.
- ▶ Three unchanged runs, then compare prediction with travel.

DONE TONIGHT Calculate turns; three distance trials; mean and range.	IF TIME REMAINS Tune one setting.
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WATCH FOR
Not documented in the companion for this week — add what you see in the room.

OPEN ITEM
This is where *error, mean and range* are core. If the optional table stays on the Week 1 page, students will have guessed their way to 100 cm before being taught to calculate it.

Repeat and turn

Mission: Use a loop to drive a square.

STUDENT NOTEBOOK
pp. 16–17

PROGRAMMING GUIDE
p. 3

MENTOR COMPANION
p. 8

BEFORE STUDENTS ARRIVE
Nothing new.

PORTS TONIGHT
B/C drives

THE HOUR — 50 PLANNED, 10 HELD FOR RECOVERY

5 arrive • 7 predict • 25 build & test • 8 evidence • 5 pack up

Teaching notes

- ▶ Calibrate first: repeat the turn 3 times unchanged, adjust M until the heading is near 90°, record mean and range (M p. 8).
- ▶ Only then wrap it in the counted loop from Guide p. 3.
- ▶ The turn is what decides whether the shape closes — not the straight runs.

DONE TONIGHT

Calibrate 90° with three turns; build the loop; one square.

IF TIME REMAINS

Complete three squares; polygon.

WATCH FOR

Not documented in the companion for this week — add what you see in the room.

Feel a hit

Mission: Detect contact. Stop and back away once.

STUDENT NOTEBOOK

pp. 18–19

PROGRAMMING GUIDE

pp. 4, 7

MENTOR COMPANION

p. 9

BEFORE STUDENTS ARRIVE

Touch sensor mount parts prepared. Check the mount coexists with the final robot and keeps moving parts clear (M p. 2).

PORTS TONIGHT

1 = touch

THE HOUR — 50 PLANNED, 10 HELD FOR RECOVERY

5 arrive • 7 predict • 8 mount • 17 build & test • 8 evidence • 5 pack up

Teaching notes

- **Mount first, then teach the condition** (M p. 9). Reserve the 8 minutes before any code.
- Teach the condition motors-off first, then add the approach.
- Guide p. 4 is if/else + repeat; Guide p. 7 adds variables and the timeout clock.

DONE TONIGHT

Mount touch; beep condition; approach with contact and timeout.

IF TIME REMAINS

Three valid gap trials.

WATCH FOR

Record the contact result, the no-contact timeout result, and the actual values used (M p. 9).

Can it stay in?

Mission: Stay inside the ring for 60 seconds.

STUDENT NOTEBOOK
pp. 20–21

PROGRAMMING GUIDE
p. 5

MENTOR COMPANION
p. 10

BEFORE STUDENTS ARRIVE

Downward-facing colour sensor mount parts prepared.

PORTS TONIGHT

3 = reflected-light colour

THE HOUR — 50 PLANNED, 10 HELD FOR RECOVERY

5 arrive · 7 predict · 8 mount · 17 build & test · 8 evidence · 5 pack up

Teaching notes

- Reserve 8 minutes for the downward mount, port 3 (M p. 10).
- Take the ten readings before choosing anything — the threshold comes from their own numbers.
- Teach it as a *repeated* comparison inside a loop, not a single check (M p. 10).

DONE TONIGHT

Mount light; five floor and five tape readings; show one recovery.

IF TIME REMAINS

Three bounded trials; blind spots.

WATCH FOR

Acceptance and limits are on M p. 10 — read them before students rely on the threshold.

Find the opponent

Mission: Find a target with sound. Keep motors off today.

STUDENT NOTEBOOK
pp. 22–23

PROGRAMMING GUIDE
pp. 5, 8

MENTOR COMPANION
p. 11

BEFORE STUDENTS ARRIVE

Forward-facing ultrasonic mount parts prepared.

PORTS TONIGHT

4 = ultrasonic cm

THE HOUR — 50 PLANNED, 10 HELD FOR RECOVERY

5 arrive • 7 predict • 8 mount • 17 build & test • 8 evidence • 5 pack up

Teaching notes

- ▶ **Motors stay stopped all session** (M p. 11). This is a sensing lesson, not a driving one.
- ▶ Three readings each at 10, 30 and 60 cm against a known target, then test an angled target and a background.
- ▶ A distance is a clue, not an identity — the sensor cannot tell a robot from a wall.

DONE TONIGHT

Mount distance; known-target readings; near-interval beep.

IF TIME REMAINS

Angles, background, interference.

WATCH FOR

A detected fault can stop motion; a plausible but wrong reading may go undetected (M p. 1). Have students name one reading they would not trust.

Choose what happens

Mission: Combine your robot's decisions in one loop.

STUDENT NOTEBOOK

pp. 24–25

PROGRAMMING GUIDE

pp. 6, 8, 9, 10

MENTOR COMPANION

pp. 12–14

BEFORE STUDENTS ARRIVE

Confirm Bevel Bot A drive, the Week 4 B/C base, and secure sensor mounts before students use anything (M p. 14).

PORTS TONIGHT

1 = touch · 3 = colour · 4 = ultrasonic

THE HOUR — 50 PLANNED, 10 HELD FOR RECOVERY

5 recap · 10 variables & motors-off trace · 20 guided assembly · 10 tests · 5 save

Teaching notes

- Build from tested pieces only. Teach Guide pp. 7–10 **in stages** (M p. 13).
- Students choose their own calibrated thresholds and explain their priority order.
- A state remembers which action is running — nested if/else plus variables (M p. 13).
- Start with motors-off readouts before anything drives (M p. 14).

DONE TONIGHT

Guided build stages; edge priority, timeout and contact stop demonstrated.

IF TIME REMAINS

Full transition-case retests.

WATCH FOR

Strategy transition checks are on M p. 14. This is the heaviest session of the term — the 20-minute assembly block is guided, not independent.

Make one improvement

Mission: Make one change. Show whether it helped.

STUDENT NOTEBOOK
pp. 26–27

PROGRAMMING GUIDE
— no new code

MENTOR COMPANION
p. 21 (concept notes) • p. 14

BEFORE STUDENTS ARRIVE

Nothing new. Teams work on their own robots.

PORTS TONIGHT

unchanged

THE HOUR — 50 PLANNED, 10 HELD FOR RECOVERY

5 arrive • 7 predict • 25 build & test • 8 evidence • 5 pack up

Teaching notes

- Choose one metric and a target, record three baseline runs, change **one** thing, repeat three times.
- Compare both means against the target and both ranges. Keep or undo, and say why.
- M p. 21 covers keeping a comparison fair and what mean, range and error each answer.

DONE TONIGHT

One metric; three before and three after trials; defend the change.

IF TIME REMAINS

Peer repeat; final inspection.

WATCH FOR

Not documented in the companion for this week — add what you see in the room.

OPEN ITEM

Two open items on this week. It is the only week with no dedicated mentor page — p. 21 is concept notes, not a session recipe. And the public parent syllabus describes Week 11 as “Design the edge — wedges, plows and low profiles,” which is a different session from this one. Resolve before the term starts.

Tournament day

Mission: Welcome your family. Compete, learn, and explain your design.

STUDENT NOTEBOOK

pp. 28–29 • published rules pp. 30–31

PROGRAMMING GUIDE

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MENTOR COMPANION

p. 15

BEFORE STUDENTS ARRIVE

Two identical rings: flat pale floor, 120 cm outer.
Student pp. 30–31 are the published camp rules (M p. 15).

PORTS TONIGHT

unchanged

THE HOUR — 50 PLANNED, 10 HELD FOR RECOVERY

Proposed — 10 welcome & inspect • 30 pool matches • 5 explain • 5 reflection • 10 spare

Teaching notes

- ▶ Choose complete round-robin groups — points are only compared within a completed group (M p. 15).
- ▶ Inspection and rulings, then the design award: M p. 15.
- ▶ Every student explains one design choice with evidence, not opinion.

DONE TONIGHT

Family pool matches; design explanation; reflection.

IF TIME REMAINS

Friendly extra matches.

WATCH FOR

Not documented in the companion for this week — add what you see in the room.

OPEN ITEM

The companion gives pacing for eleven weeks and stops. The clock above is a proposal built from the notebook's running order and the 50 + 10 frame — replace it with the real format.